Hotazel Steam: Revenue Forecasting Business Brief

Business Context

Hotazel Steam is a public utility based in New England that produces steam used for both heating in winter and cooling in summer. Because demand for steam is driven by weather conditions, its monthly revenues fluctuate significantly throughout the year. Management needs reliable revenue expectations to support operational planning, budgeting, and financial reporting, but predicting revenue is difficult when it depends on factors outside the company's control like temperature, gas prices, and broader energy trends.

QuantFolio Solutions was engaged to build and evaluate regression models that predict monthly revenue using available operational and weather data. The data spans January 2011 through December 2014, split so that 2011 to 2013 serves as the training period and 2014 serves as the testing period to evaluate how well each model predicts revenue it has never seen before.

Week 8 Notebook: Simple Linear Regression

This notebook builds the first generation of Hotazel Steam forecasting models using simple linear regression. Each model uses a single independent variable, either production volume or a weather variable like coolDD or heatDD, to predict monthly revenue. The goal is to identify which single variable best explains revenue and establish a baseline model to build on.

Week 10 Notebook: Dummy Variables and Interaction Terms

This notebook extends the Week 8 models by introducing dummy variables and interaction terms to capture Hotazel Steam's seasonal revenue patterns. A winter dummy (November through January) and a summer dummy (June through August) allow the model's intercept to shift by season, reflecting the reality that heating and cooling demand create fundamentally different revenue baselines throughout the year. Interaction terms further allow the slope on production to vary by season. The goal is to find the combination of variables that produces the lowest MAPE and highest adjusted R-squared, giving management the most accurate revenue forecasting tool available.